

VIMD-Net: Physically Supervised Representation Allocation for Automatic Modulation Classification (AMC) under Vehicular-Doppler TDL Simulations

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Abstract—Automatic modulation classification (AMC) under mobile multipath is impaired by structured interference that overlaps modulation cues in time and frequency. This paper develops VIMD-Net, a representation-allocation network whose environment-conditioned three-way real mask acts on complex short-time Fourier transform coefficients and allocates target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous evidence. A fixed training-only teacher is constructed from independently tracked target, jammer, and unexplained simulation components; dual task branches, auxiliary jammer/link supervision, and exact-source cross-condition contrastive learning constrain the route semantics. The evaluation is locked to source-disjoint 3GPP TR 38.901 TDL-profile simulations parameterized by vehicular Doppler and structured jammers, with independently held-out jammer, speed, and channel factors, single-view inference, and source-paired statistical comparisons. Unanchored in-taxonomy modulated cochannel collisions are excluded because their single-label target is not identifiable from a mixture alone. Quantitative claims are intentionally withheld in this internal-review build until the frozen multi-seed evidence gate is passed.

Index Terms—Automatic modulation classification, vehicular communications, interference robustness, representation disentanglement, soft masks, contrastive learning.

Evidence lock. This build is for internal scientific review. Em-dash cells are not zeroes and must not be interpreted as experimental outcomes. Smoke runs and fixed-batch overfit diagnostics are excluded from paper claims.

I. INTRODUCTION

AUTOMATIC modulation classification enables a receiver to infer the modulation of an unknown waveform without transmitter-side control information. It supports spectrum monitoring, blind receiver configuration, interference diagnosis, and adaptive signal processing. Deep AMC models have advanced from convolutional classifiers [1], [2] to complex-valued [3], dual-stream [4], causal-convolutional [5], and Transformer-based [6], [7] representations. Efficiency, hardware impairment, and channel robustness have also been studied [8]–[10].

A mobile receiver poses a different failure mode from the usual additive-white-Gaussian-noise benchmark. Mobility produces time-varying multipath and Doppler, while concurrent transmitters and unintended emitters create narrowband, impulsive, swept, partial-band, and broadband interference. The target and jammer can therefore share both time and frequency support. Direct classifiers must learn modulation

and nuisance factors from a single polluted representation. A hard suppression stage is also unsafe: a cell with high jammer power may still contain target power.

Recent work narrows individual parts of this gap. Domain adaptation improves cross-domain AMC [11], [12]; masked and contrastive learning improves representation robustness [13], [14]; interference-tolerant feature extraction directly addresses corrupted observations [15]; and overlapped-signal classification studies simultaneous emitters [16]. Consequently, neither “soft masking,” “contrastive AMC,” nor “multi-task AMC” is a defensible novelty claim by itself. What remains insufficiently tested is whether a mask can be assigned an auditable physical meaning, explicitly retain ambiguous target–jammer cells, and improve modulation invariance under independently held-out vehicular and jammer conditions.

This work addresses that question with VIMD-Net. It is derived from a broader interference-adaptation concept, but replaces template matching, binary component deletion, and rule-based label correction with differentiable, task-constrained representation allocation. Its positive claim is deliberately narrow: it tests whether component-power supervision improves AMC under source-disjoint structured-interference and TDL-profile simulations.

The contributions are threefold.

- We introduce an environment-conditioned three-way mask on the exact complex-STFT lattice. It allocates target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous evidence with a learned overlap gate and a bounded additive bypass.
- We construct a fixed, component-traceable simulation-component power teacher from target, jammer, and unexplained power on that same lattice. Jensen–Shannon mask supervision and separate modulation, jammer, and link-quality objectives make the branch semantics experimentally falsifiable.
- During training, we pair two independently impaired observations of the same immutable source sequence and apply class-aware exact-source contrastive learning; inference uses one view. The accompanying protocol isolates jammer-family, speed, and channel-profile shifts and pre-declares accuracy, calibration, source-paired significance, diagnostic mechanism variables, and complexity.

II. RELATED WORK AND CLAIM BOUNDARY

A. Deep AMC

CNN-based AMC established that learned I/Q features can outperform hand-designed pipelines under matched synthetic

conditions [1], [2], [17]. Later work preserved complex structure [3], fused spatial and temporal streams [4], reduced complexity [8], and improved causal or Transformer modeling [5]–[7]. These models form necessary direct-classification comparators; architectural depth alone cannot establish interference disentanglement.

B. Robustness, Masking, and Contrastive Learning

Channel-robust hand-crafted statistics [10], explicit hardware impairment compensation [9], domain adaptation [11], [12], masked contrastive learning [13], denoising masked autoencoding [18], and a recent IEEE contrastive learner [14] address different nuisance sources. MFENet explicitly targets interference tolerance [15]; malicious-interference recognition has also been studied for MIMO observations [19].

Time–frequency masks are established speech-separation targets rather than a new primitive of this work. Prior studies compared ideal binary and ratio masks [20], [21], introduced phase-sensitive masks [22], estimated complex ratio masks that can change phase [23], and learned source partitions with deep clustering [24]. VIMD-Net transfers the component-traceable masking idea to AMC, but its teacher differs from a two-source ratio target: the third route retains receiver/noise power and target–jammer equal-dominance mass rather than discarding ambiguous cells.

Our narrow distinction is therefore a three-route component-power teacher on the student’s physical lattice, with the other physical condition of the exact source as the contrastive positive. Its target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous names are task assignments, not identified sources.

C. From Patent Concept to Scientific Hypothesis

The source concept follows an environment adaptation → interference suppression → modulation decision chain. The scientific implementation makes four material changes. First, a continuous environment embedding replaces a discrete jammer template database. Second, soft three-way allocation replaces hard threshold deletion. Third, a discriminative classifier replaces nearest-prototype and rule-forced label revision. Fourth, the jammer state is an auxiliary output and never overrides the observed modulation class. This separation prevents a link-adaptation recommendation prior from being confused with signal identification.

III. SYSTEM MODEL AND LEARNING PROBLEM

A. Received Signal

For a length- N complex baseband window, let

$$x[n] = \mathcal{G}_\eta \left(\begin{array}{l} \sum_{\ell=0}^{L_s-1} h_s[n, \ell] s_m[n - \ell] \\ + \sum_{q=1}^Q a_q \sum_{\ell=0}^{L_{j,q}-1} h_{j,q}[n, \ell] j_q[n - \ell - \tau_q] \\ + w[n] \end{array} \right), \quad (1)$$

where s_m is a source sequence with modulation label m , h_s and $h_{j,q}$ are independently seeded time-varying channels, j_q is a structured interferer, and $\mathcal{G}_\eta$ collects receiver effects shared by the complete observation. Equation (1) permits a general Q , but the present evidence protocol admits exactly one structured jammer family per active view; multi-jammer compositional mixtures are outside scope. The generator retains the post-channel components

$$x = s + j + u, \quad u = w + a_{\text{rx}}, \quad (2)$$

after a common gain convention, where a_{rx} is the tracked receiver artifact. Realized powers, rather than requested scale factors, define

$$\text{SNR} = 10 \log_{10} \frac{\|s\|_2^2}{\|w\|_2^2}, \quad \text{SIR} = 10 \log_{10} \frac{\|s\|_2^2}{\|j\|_2^2}. \quad (3)$$

No-jammer samples carry a validity mask instead of a finite fictitious SIR.

The main task estimates $p_\theta(m | x)$. Training-only auxiliary targets are a multi-hot jammer vector and normalized SNR, SIR, and maximum Doppler. At inference, only x is required.

B. Exact-Source Paired Views

An immutable source identity contains the modulation, symbol sequence, and pulse-shaping realization. For source i , two views are

$$x_i^{(v)} = \mathcal{A} \left(s_i; h_{s,i}^{(v)}, h_{j,i}^{(v)}, j_i^{(v)}, \eta_i^{(v)} \right), \quad v \in \{1, 2\}, \quad (4)$$

with independent channel, jammer, receiver, and noise seeds. All descendants of one source identity remain in one split. Thus the pair supplies a valid training positive without allowing bit-sequence leakage across train, validation, or test sets. Only one received view is supplied for each inference decision; the second view is used during paired training and source-aligned statistical comparison, not as test-time side information.

IV. VIMD-NET

A. Physically Aligned Spectral Front End

The input is represented as $X = [I; Q] \in \mathbb{R}^{2 \times N}$. A deterministic, non-centered, full-spectrum complex STFT produces

$$Z = \text{STFT}(I + jQ) \in \mathbb{C}^{F \times T}. \quad (5)$$

The student context encoder receives

$$H = C_\phi(\text{stack}\{\Re Z, \Im Z, \log(1 + |Z|)\}), \quad (6)$$

where C_ϕ is a shallow 2-D convolutional residual encoder that preserves the $F \times T$ mask lattice. The real-valued encoder receives real, imaginary, and log-magnitude planes; it is not an end-to-end complex-valued neural network.

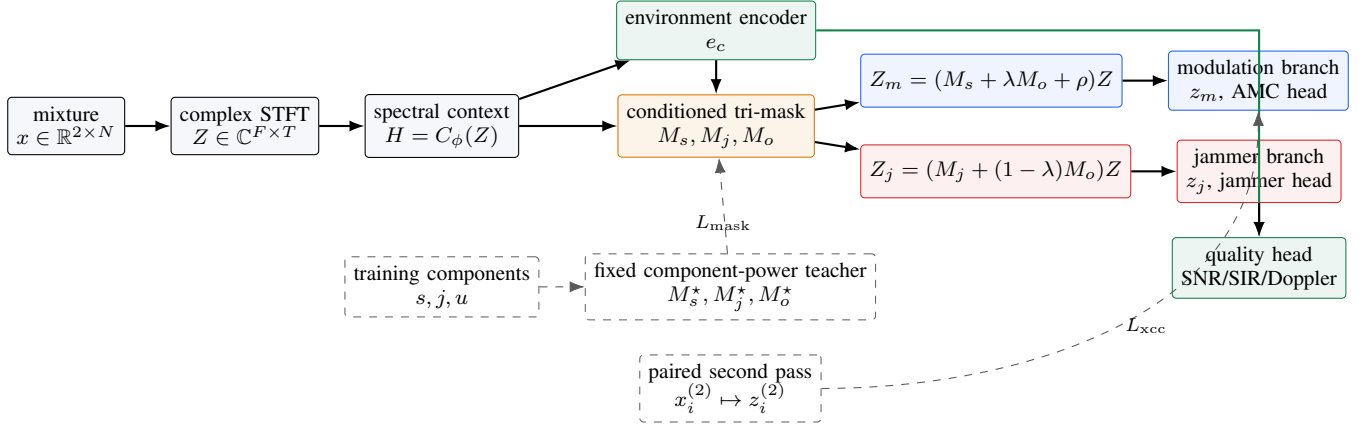


Fig. 1. Implemented VIMD-Net data flow. Solid paths are used at inference; dashed paths exist only during training. The masks act on the complex STFT itself, so the teacher and student share the same frequency-frame cells. The bounded additive ρ bypass intentionally makes the two branch weights non-conservative; whether it improves target-task robustness is tested by the predeclared A5–A7 comparison.

B. Environment-Conditioned Tri-Mask

Mean, standard deviation, and maximum pooling form

$$\text{Stats}(H) = [\mu(H), \sigma(H), \max(H)], \quad (7)$$

$$e_c = E_\psi(\text{Stats}(H)). \quad (8)$$

Feature-wise conditioning is

$$\tilde{H} = (1 + \frac{1}{2} \tanh \gamma(e_c)) \odot H + \beta(e_c). \quad (9)$$

The model predicts

$$\tau_c = \tau_{\min} + (\tau_{\max} - \tau_{\min}) \sigma(g_\tau(e_c)), \quad (10)$$

$$[M_s, M_j, M_o] = \text{softmax}_{\text{route}}(G_\omega(\tilde{H})/\tau_c), \quad (11)$$

so the three nonnegative masks sum to one at every time-frequency cell. Here M_s , M_j , and M_o denote target-power-dominant, jammer-power-dominant, and unexplained-or-power-ambiguous allocations, respectively. These are component-power task routes, not identified sources.

Two bounded gates allocate the ambiguous route:

$$\lambda = \sigma(g_\lambda(e_c)), \quad (12)$$

$$\rho = \rho_{\min} + (\rho_{\max} - \rho_{\min}) \sigma(g_\rho(e_c)), \quad (13)$$

$$Z_m = (M_s + \lambda M_o + \rho) \odot Z, \quad (14)$$

$$Z_j = (M_j + (1 - \lambda) M_o) \odot Z. \quad (15)$$

Importantly, λ , ρ , and τ_c are condition-dependent but are one scalar per received window. This low-dimensional choice makes condition-level allocation auditable and avoids a second unconstrained time-frequency gate, but it cannot vary the ambiguous-route split within a sweep or pulse. A per-cell $\lambda[f, t]$ sensitivity is therefore a distinct unexecuted variant, and no within-window adaptivity is attributed to the scalar gate. The gates are not constrained to be monotonic in SIR; any such relationship must be measured rather than asserted.

C. Fixed Simulation-Component Three-Route Teacher

Let $P_s = |\text{STFT}(s)|^2$, $P_j = |\text{STFT}(j)|^2$, and $P_u = |\text{STFT}(u)|^2$ on the same lattice as Z . Define

$$q_k = \frac{P_k}{P_\Sigma}, \quad P_\Sigma = P_s + P_j + P_u > 0, \quad k \in \{s, j, u\}, \quad (16)$$

The teacher has the direct cellwise form

$$M_s^* = [q_s - q_j]_+, \quad (17)$$

$$M_j^* = [q_j - q_s]_+, \quad (18)$$

$$M_o^* = q_u + 2 \min(q_s, q_j). \quad (19)$$

Thus equal target-jammer power is assigned to uncertainty, while receiver artifacts and noise are always unexplained. The teacher is deterministic, has no trainable or exponentially averaged parameters, and is removed at inference. A zero-power cell is assigned $(0, 0, 1)$. The implementation uses the per-example guard $\epsilon_p = \max(10^{-8} \text{mean}_{f,t} P_\Sigma, \text{tiny})$: cells with $P_\Sigma \leq \epsilon_p$ are assigned $(0, 0, 1)$, all other denominators are lower-bounded for numerical safety, and every cell is finally renormalized to the simplex. The finite-precision implementation deliberately maps negligible-energy cells to uncertainty.

a) *Algebraic properties.*: The three nonnegative quantities in (17)–(19) sum to one because $[q_s - q_j]_+ + [q_j - q_s]_+ + 2 \min(q_s, q_j) = q_s + q_j$. They are also unchanged by a common nonzero complex scaling of all three components because the scale cancels in every q_k . These are direct properties of the definition, not theoretical performance claims.

For completeness, multiplication by a real nonnegative weight satisfies $\angle(W[f, t]Z[f, t]) = \angle Z[f, t]$ whenever $W[f, t] > 0$; at zero the coefficient is suppressed and its phase is undefined. This elementary identity is shared by every real magnitude mask and is not a contribution. Unlike phase-sensitive [22] or complex-ratio [23] masking, VIMD-Net estimates no phase correction: each nonzero branch coefficient retains the mixture STFT phase.

D. Dual Branches and Exact-Source Contrast

Independent spectral encoders yield $z_m = P_m(Z_m)$ and $z_j = P_j(Z_j)$. A linear head predicts modulation from z_m ; a multi-label head predicts jammer presence from z_j ; and a quality head predicts normalized SNR, valid SIR, and Doppler from $[e_c, z_j]$.

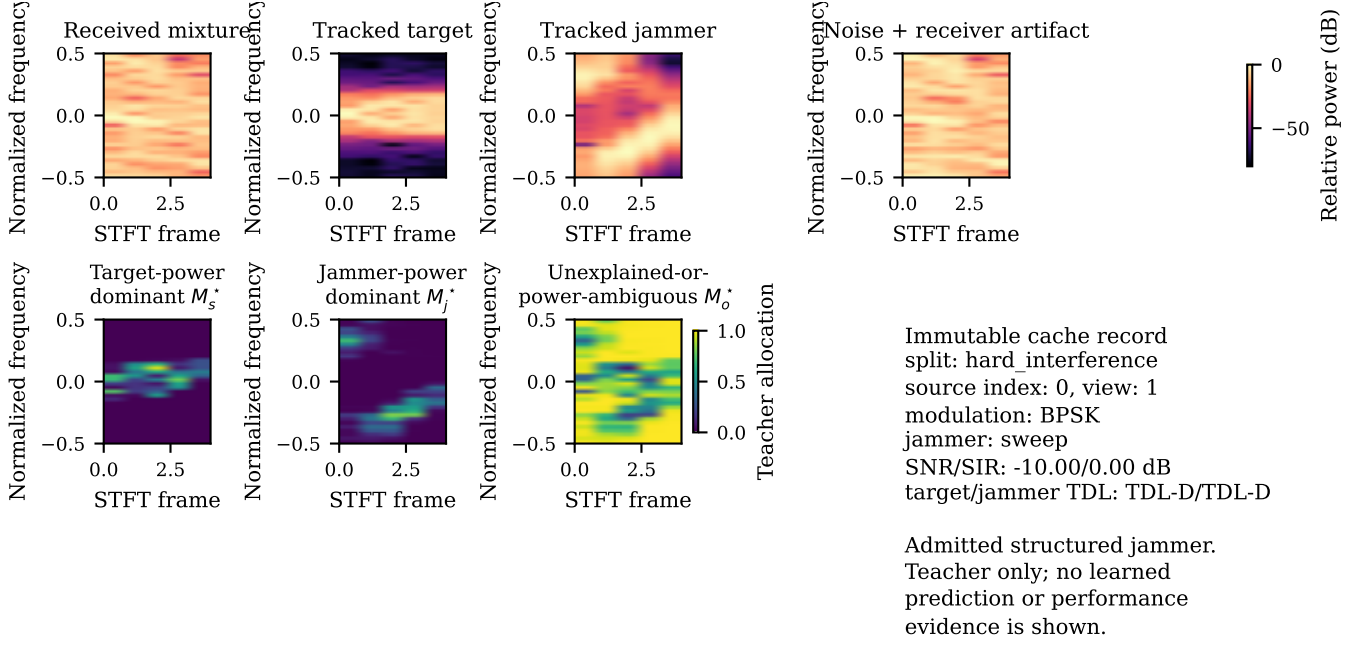


Fig. 2. Fixed-teacher proxy visualization from one immutable *micro-smoke* hard-interference cache record containing an admitted structured sweep jammer. It uses $L = 64$ and STFT 32/8, hence five frames; the formal protocol uses $L = 1024$ and STFT 64/16, hence 61 frames. The upper row shows the mixture and tracked post-channel simulation components, and the lower row shows allocations on that same proxy lattice. This figure explains the definition only; it is neither a learned prediction nor performance evidence.

For a batch of paired views, let $\tilde{z}_i^{(1)}$ and $\tilde{z}_i^{(2)}$ be unit-normalized modulation embeddings. The valid comparison set for anchor i is

$$\mathcal{A}_i = \{i\} \cup \{k : y_k \neq y_i\}. \quad (20)$$

The diagonal is the other view of the exact same source; other sources of the same modulation are ignored. One direction of the loss is

$$\ell_i^{1 \rightarrow 2} = -\log \frac{\exp(\tilde{z}_i^{(1)\top} \tilde{z}_i^{(2)} / \tau_x)}{\sum_{k \in \mathcal{A}_i} \exp(\tilde{z}_i^{(1)\top} \tilde{z}_k^{(2)} / \tau_x)}. \quad (21)$$

The cross-condition loss averages both directions. Unlike generic supervised contrastive learning [25], it neither treats all same-class signals as positives nor turns a different source of the same class into a false negative. The current method does not claim jammer- or SIR-aware hard negative mining.

E. Objective and Curriculum

The full objective is

$$\mathcal{L} = L_{\text{mod}} + \lambda_j L_{\text{jam}} + \lambda_q L_q + \lambda_m L_{\text{mask}} + \lambda_x L_{\text{xcc}} + \lambda_{\perp} L_{\perp}, \quad (22)$$

where $L_{\text{mask}} = D_{\text{JS}}(M^* \| M)$, L_q is validity-masked smooth L_1 , and $L_{\perp}$ is the squared centered cosine similarity between active jammer-task and modulation-task embeddings. This orthogonality term is evaluated only for jammer-active views and does not establish source identifiability. The locked initial weights are $(\lambda_j, \lambda_q, \lambda_m, \lambda_x, \lambda_{\perp}) = (0.25, 0.05, 0.50, 0.10, 0.01)$ with label smoothing 0.05 and $\tau_x = 0.12$. Mask and contrastive terms ramp in after the classification warm-up. Learning-rate decay and checkpoint

selection begin only after the complete objective has been active for a minimum number of epochs, preventing early validation checkpoints from bypassing the proposed losses. Natural-log Jensen–Shannon divergence is symmetric, finite for zero-probability teacher routes, and bounded by $\log 2$ [26]. Those support and scale properties motivate it over a directed KL loss; boundedness can also weaken gradients when student and teacher are far apart, so teacher agreement and loss curves remain required diagnostics rather than assumed consequences of the objective.

V. SIMULATION EVIDENCE PROTOCOL

A. Data Layers

The evidence is intentionally layered.

- 1) A deterministic Python proxy generator is used for unit tests, diagnostics, and inexpensive screening. Its channel heuristics cannot support a standards-compliance claim.
- 2) The frozen headline-cache builder is configured to use MATLAB `nrTDLChannel` primitives corresponding to TDL-A through TDL-E in 3GPP TR 38.901 [27]. Target and jammer waveforms traverse independent channel objects before shared receiver effects and exact realized SNR/SIR scaling. TDL-A/C/D are allocated to training and validation; TDL-B/E are reserved for a profile-held-out test.

TDL profiles alone do not instantiate every geometry, block-age, or nonstationary trajectory in a V2X evaluation methodology [28]. We therefore use the narrower term “TR 38.901 TDL-profile simulations.” The complete claim boundary is consolidated in Section VIII.

B. Signal, Jammer, and Impairment Space

The locked target set contains BPSK, $\pi/2$ -BPSK, QPSK, 8PSK, 16QAM, 64QAM, 256QAM, GMSK, CPFSK, and 4FSK. The seen jammer dictionary contains tone, multitone, chirp, sweep, partial-band, and comb waveforms; pulse and OFDM-like interference are held out by family. Modulated cochannel collisions and multi-jammer compositional mixtures are excluded from the single-label headline protocol. Here, *cochannel* means an in-taxonomy modulated cochannel collision, whereas *mixed* means a composition of multiple jammer families. Without a target anchor, swapping two in-taxonomy cochannel emitters can leave the observation distribution unchanged while changing the requested label. Only on a balanced, exchangeable source class with identical x and two required labels is the mixture-only Bayes error at least $1/2$; the bound is not asserted for unequal received powers or distinguishable channel statistics. Such collisions require a dominant-emitter label, an explicit physical anchor, or a separately scored ambiguity task. A practical dominant-emitter cochannel split with a preregistered received-power margin of at least 6 dB remains unexecuted and is outside the present headline protocol. The exact discrete SNR grid for every split is $\{-10, -6, -2, 2, 6, 10, 14, 18\}$ dB. Jammer-active ordinary splits use $\text{SIR} \in \{-10, -5, 0, 5, 10\}$ dB, while the hard-interference split uses $\text{SIR} \in \{-15, -10, -5, 0\}$ dB. Clean samples have invalid SIR by construction. Training samples are drawn only from these declared discrete grids; there is no continuous training interpolation.

Nominal receiver factors include carrier-frequency and phase offset, I/Q gain/phase imbalance, additive DC offset, and window-level AGC normalization. The shared multiplicative effects preserve the component audit in (2), while DC is tracked in the unexplained component. Three test-only robustness splits extend this nominal model. Two quantize the post-AGC real and imaginary samples with a symmetric 10- or 12-bit ADC at a fixed full scale; clipping incidence and the exact quantization residual are recorded, and the residual is assigned to the unexplained component. The third independently draws target and jammer carrier offsets in $[-0.006, 0.006]$ normalized frequency and integer timing offsets in $[-3, 3]$ samples on their separately channelized pre-mixing branches. Sampling-clock drift and additional receiver filtering remain outside the evidence scope.

C. Factor-Isolated Splits

Table I prevents a combined shift from hiding its cause. Every split has independent source identities and generator/channel seeds. Training, validation, and ID testing use a deterministic 20% no-jammer quota; every stress split is jammer-active, whereas clean retention is entirely jammer-free. The combined set is a stress test, not a substitute for the three isolated tests.

D. Cache Audit

The offline builder pre-appends a fixed guard, batches MATLAB channel calls, audits returned filter and path delays,

TABLE I
LOCKED PROTOCOL SPLITS. “HELD” MEANS ABSENT FROM TRAINING.

Split	Jammer/composition	Speed	TDL profile
Train	seen + 20% clean	seen	A/C/D
Validation	seen + 20% clean	seen	A/C/D
ID test	seen + 20% clean	seen	A/C/D
Hard interference	seen, $\text{SIR} \leq 0$	seen	A/C/D
Unseen jammer	held	seen	A/C/D
Unseen speed	seen	180/250 km/h	A/C/D
Held-out channel	seen	seen	B/E
Combined OOD	held	180/250 km/h	B/E
Clean retention	none	seen	A/C/D/B/E
ADC 10-bit + AGC	seen, $\text{SIR} \leq 0$	seen	A/C/D
ADC 12-bit + AGC	seen, $\text{SIR} \leq 0$	seen	A/C/D
Per-emitter CFO/timing	seen, $\text{SIR} \leq 0$	seen	A/C/D

TABLE II
REQUIRED FAIL-CLOSED CACHE-AUDIT GATES FOR AN ELIGIBLE HEADLINE MANIFEST; NO SCREENING OR INTEGRATION SENTINEL IS PROMOTED.

Gate	Required manifest evidence
Component identity	reload residual and tolerance
Power realization	SNR/SIR errors and validity masks
Guard adequacy	requested and post-delay margins
Immutability	file checksums and manifest digest
Source disjointness	pair/split source-ID intersections

then crops the central window. It records target and jammer profile, delay spread, maximum Doppler, channel seed, guard margin, source ID, requested and realized SNR/SIR, component residual, file SHA-256, and a deterministic manifest digest. If built and validated, the eligible headline manifest is the sole source of reportable audit values. The current screening and integration sentinels validate pipeline structure only and are not performance evidence.

VI. EXPERIMENTAL DESIGN

A. Frozen Implementation and Protocol

Table III records the prospective v2 formal configuration, which supersedes v1 for future execution without reinterpreting any historical v1 output. The formal headline cache and run have not been generated in this internal-review build; only screening/integration cache evidence exists, and none of its metrics enters a result table.

B. Comparators and Ablations

All trainable models receive identical source pairs, split manifests, epoch budget, optimizer family, label smoothing, and checkpoint policy. Literature models are labelled “reimplementation,” “inspired,” or “supervised adaptation” according to their provenance unless official code and the complete published protocol are executed unchanged. Quoted paper numbers are never mixed with unified retraining.

The A2–A5 chain adds one predeclared intervention block at each step under the shared protocol: the fixed teacher at A3–A2, the bundled jammer, quality, and orthogonality objectives at A4–A3, and exact-source contrast at A5–A4. The required teacher-form control is A3’, runner `a3p_tri_proportional_teacher` (alias

TABLE III
FROZEN IMPLEMENTATION AND FORMAL EVIDENCE PROTOCOL. VALUES ARE CONFIGURED PROSPECTIVELY; ELIGIBILITY REQUIRES AN
ARTIFACT-BOUND RELEASE LOCK.

Item	Frozen setting	Item	Frozen setting
Input/cache	$L = 1024$ complex I/Q samples; 96-sample guard; $f_s = 1$ MHz; $f_c = 5.9$ GHz; master seed 20260727	Targets	BPSK, $\pi/2$ -BPSK, QPSK, 8PSK, 16/64/256QAM, GMSK, CPFSK, and 4FSK
Linear waveforms	Fixed 4 samples/symbol ($R_s = 250$ ksym/s); energy-normalized RRC, roll-off 0.35, span 8 symbols; integer timing phase 0, $\dots$, 3 randomized per source	Continuous phase	GMSK: $h = 0.5$, Gaussian $\sigma = 0.55T_s$, truncated to $\pm 4T_s$ (nominal $BT \approx 0.24$); CPFSK: $h = 0.5$; 4FSK offsets $\{-0.36, -0.12, 0.12, 0.36\}R_s$. Parameters fixed; data and initial phase randomized
Channel/mobility	TDL-A/C/D seen, TDL-B/E held; 30/100/300-ns delay spreads; 0/60/120/150 km/h seen, 180/250 km/h held	Jammers	tone, multitone, chirp, sweep, partial-band, and comb seen; pulse and OFDM-like held; cochannel and mixed excluded
Quality grids	SNR $\{-10, -6, -2, 2, 6, 10, 14, 18\}$ dB; ordinary SIR $\{-10, -5, 0, 5, 10\}$ dB; hard SIR $\{-15, -10, -5, 0\}$ dB	Source budget	train/validation 100,000/2,000; each of ten test splits 5,000; 152,000 disjoint source IDs and 304,000 cached views; pre-registered A0/A5 scale check at 10k/30k/100k precedes the confirmatory run, with 100k fixed regardless of that curve
Transform/widths	Periodic-Hann STFT 64/16 (FFT/hop), center=false, unitary normalization, full two-sided spectrum, 61 frames; complex widths 8/16; feature/spectral widths 32/24; embedding/environment 48/32; dropout 0.05	Routing/objective	$\rho \in [0.05, 0.35]$, $\tau_c \in [0.5, 2]$; $\lambda_{j,q,m,x,\perp} = (0.25, 0.05, 0.50, 0.10, 0.01)$; smoothing 0.05; $\tau_x = 0.12$
Optimization	AdamW; 30 epochs; batch 64; $\eta = 3 \times 10^{-4}$; weight decay 10^{-2} ; gradient clip 5; workers 0; CPU intra/inter-op threads 1/1; CUDA AMP (teacher in FP32)	Curriculum/selection	mask/XCC first active at epochs 3/6 (one-based), each ramps for 3 epochs; full objective at epoch 8; selection starts at 10; patience 8
Execution matrix	A0–A7 plus A3', MCLDNN, IQFormer-inspired, and CSSL adaptation; seeds $\{17, 29, 43, 71, 101, 131, 173, 211, 257, 307\}$; 12 models $\times$ 10 seeds = 120 confirmatory fits; CSSL-AMC official-architecture supervised adaptation is the fixed primary anchor	Inference/statistics	one view; 10,000-draw seed/source hierarchical paired bootstrap (seed 20260803, distinct from the generator seed); class-stratified source clusters; validation excluded
Latency protocol	batch one; 10 warm-up passes; 30 synchronized timed passes; P50/P95 on the recorded isolated device	Release integrity	checksum and component validation; unchanged source tree; eligible checkpoint, no fallback; artifact-derived result cells only

a3_ratio_teacher): it replaces the margin teacher by the direct proportions (q_s, q_j, q_u) while holding A3 architecture and optimization fixed. The confirmatory teacher-form contrast is A3–A3'; it is prospectively registered, and the protocol assigns it no value before an eligible run. The A4–A3 block is not decomposed into three separate effects. A5–A6 is a composite control because route count, capacity, and overlap allocation change together; it does not identify a route-count-only causal effect. The complete A0–A7 numeric ladder is placed with the formal results in Table IV; in the internal build its cells remain placeholders, and public values become visible only through the eligible release lock.

The formal comparator set is exactly the A0 direct shared backbone, MCLDNN [29], the transparent IQFormer-inspired baseline, and the CSSL-AMC supervised adaptation described below. The predeclared primary reference is the CSSL-AMC adaptation; this is a fixed comparison anchor, not a post-hoc winner claim. The recent low-SNR/noise-robust AMC comparator is a 2025 CSSL-AMC official-architecture supervised adaptation [14]: its encoder/classifier graph is audited against the authors' Apache-2.0 source snapshot [30] at commit 2fbc5b3e12f7, but it is randomly initialized and trained with the common paired-view classification objective, not the published two-stage protocol. It imports neither official weights nor reported results and is not a structured-interference-specific method. MFENet [15] and RepCCNet [5] remain related work, not executed rows. For MFENet, a bounded audit found no author-controlled public code, checkpoint, or complete executable specification; this is a bounded negative search result, not evidence that no implementation exists. A classical higher-order/cyclostationary feature classifier and clean-input or oracle-teacher routes are separate non-

headline diagnostics, not formal comparators or deployable baselines. Oracle-control outcomes are prohibited from the formal release tables.

The IQFormer row is likewise labelled “inspired”: its local graph is audited against the authors' MIT-licensed repository at commit 7ee6ac949551 [31], but it is not an official-checkpoint or native-protocol reproduction. All trainable comparators receive the same two-view batches and supervised labels; the second view is therefore not an A5-only batch-composition advantage. A5–A4 changes the cross-condition loss while keeping the paired data contract fixed.

No MFENet author contact has been sent, and neither IQFormer nor CSSL-AMC has yet been reproduced under its native RadioML protocol. These are not missing inputs to the present simulation-only experiment: the manuscript labels the local IQFormer and CSSL rows as bounded adaptations, treats MFENet as related work rather than an executed comparator, and claims neither reproduction of published results nor superiority over those unexecuted methods. A future public-benchmark or broader comparative claim would require a separate prospective supplement: an auditable MFENet implementation or dated contact trail, native IQFormer/CSSL checks, and an authorized RADIOML data snapshot with immutable provenance and a frozen interference overlay. DeepSig warns that its historical open datasets have known errata [32], [33]; any such supplement must preserve that qualification. No result from that optional scope expansion is asserted here.

C. Metrics and Statistical Inference

Primary outcomes are macro-F1 and accuracy; secondary outcomes are worst supported-class recall, negative log-likelihood, and 15-bin expected calibration error. Source-

aligned prediction bundles retain per-source SNR/SIR meta-data for a separately preregistered stratified analysis. The current release reports only frozen regime aggregates and makes no released full-SNR–SIR-plane claim.

The primary reference comparator is predeclared before any model training or test evaluation and is not selected on validation. For each frozen source set, models produce source-aligned probability bundles. Candidate–reference accuracy and macro-F1 differences use a 10,000-draw, class-stratified source-cluster paired bootstrap. For the ten-seed headline comparison, the primary hierarchical paired bootstrap resamples both algorithm seeds and the same paired source clusters; source-only and seed-only intervals are also reported. An exact two-sided McNemar test is supplemental, restricted to one fitted-model seed, reported for every predeclared candidate, and never computed by pooling predictions across seeds. Holm adjustment is applied within each predeclared held-out-regime $\times$ seed family across the candidate model family; it does not correct across regimes or seeds and is not the basis for the headline inference. Validation is excluded from testing. Screening uses three model seeds, while the locked headline comparison uses ten. Before execution, the absolute macro-F1 SESOI for calling the primary A5–A0 gain practically material was fixed at 1 percentage points. Statistical success still requires its family-wise simultaneous lower bound to exceed zero; a positive interval with a point gain below 1 percentage points is reported as resolved but below the materiality threshold. The design claims no model-based power percentage because eligible paired seed/source variance estimates did not exist before the freeze; achieved simultaneous-interval width and its relation to the SESOI must therefore be reported without result-dependent seed or source extension.

D. Mechanism Tests

Accuracy alone cannot verify the proposed interpretation. Let $W_m = M_s + \lambda M_o + \rho$. For independently tracked target and jammer spectra, define

$$\alpha = \frac{\|W_m \odot Z_s\|_2^2}{\|Z_s\|_2^2}, \quad \beta = \frac{\|W_m \odot Z_j\|_2^2}{\|Z_j\|_2^2}, \quad (23)$$

$$\Delta_{\text{spec}} = 10 \log_{10} \frac{\alpha}{\beta}. \quad (24)$$

We report energy-weighted route correlation and MAE against the teacher, route occupancy, α , β , and Δ_{spec} . Because the additive bypass can make $W_m > 1$, α is a target-energy transfer ratio, not a bounded retention rate; its mean, maximum, and fraction of samples with $\alpha > 1$ are reported. Since $\rho \leq W_m \leq 1 + \rho$, nonzero target and jammer components satisfy $\beta \geq \rho^2$, $\alpha \leq (1 + \rho)^2$, and

$$\Delta_{\text{spec}} \leq 20 \log_{10} \frac{1 + \rho}{\rho} \leq 20 \log_{10} \frac{1.05}{0.05} \approx 26.4 \text{ dB}.$$

The teacher’s third route combines unexplained power and target–jammer power ambiguity, which are not identifiable as separate constituents from the student mask. Their teacher-side occupancies may be tabulated only to describe the simulated target; correlation and MAE are computed for the combined

route. A 1,000-draw permutation test evaluates the diagnostic association between predicted and teacher third-route occupancy. These mechanism probes are exploratory diagnostic associations outside the headline multiplicity family. In particular, Δ_{spec} is an oracle-conditioned spectral component ratio, not waveform SIR gain, source separation, or a real-world SIR estimator.

The auxiliary heads are evaluated directly: jammer micro/macro F1 and AUROC, and validity-masked SNR, SIR, and Doppler MAE after conversion back to physical units. Undefined AUROC or unavailable scaling metadata is reported as unavailable rather than imputed.

E. Complexity

Parameter count and serialized state size are exact. Convolution/linear MACs exclude the STFT, whose real-operation count is reported separately from the standard FFT order estimate. Batch-one P50/P95 latency is reported only for the frozen device used by the eligible run after warm-up and required device synchronization. The operator must isolate that device before measurement; the release records the device, runtime, window length, and synchronization policy. No CPU–GPU comparison or cross-device latency claim is made.

VII. RESULTS AND FALSIFICATION GATES

This section is intentionally unpopulated. The executable pipeline and screening/integration caches have passed unit, data-identity, cache, gradient-routing, and fixed-batch optimization checks, but no formal headline cache or run has been generated. The available screening profiles are too small to estimate generalization. No ineligible diagnostic or screening run is promoted into Tables IV–VII.

The method is rejected or narrowed if any of the following occurs. First, any of the three confirmatory hard-region contrasts A5–A0, A3–A3', or A5–A6 has a family-wise simultaneous 95% lower bound at or below zero. Second, an OOD axis whose A0 ID-to-axis degradation has lower 95% bound above 0.5 percentage points must have a positive lower bound for

$$(A5 - A0)_{\text{axis}} - 0.25 \max\{(A0_{\text{ID}} - A0_{\text{axis}}), 0\}.$$

If that consequential shift is not established, the axis instead requires the A5–A0 gain lower bound to be at least -1 percentage points; every OOD axis is reported and must pass its predeclared branch. The receiver-stress axes use the same branches but compare against the nominal hard-interference split, not ID, so receiver degradation is not confounded with the SIR distribution. Third, clean-condition macro-F1 degradation may not exceed 1.0 percentage points in point estimate and its paired 95% lower confidence bound may not fall below -2.0 percentage points. Clean retention is reported separately for seen-profile A/C/D and held-profile B/E strata. The six-family occupancy/gain directional test must be reported; a null or opposite result blocks a positive mechanism claim but does not block an honestly scoped submission. Supplemental McNemar tests cannot rescue any failed branch. Failures are reported rather than repaired by deleting difficult classes or conditions.

TABLE IV

ARTIFACT-LOCKED ABLATION EVIDENCE ON HARD INTERFERENCE. MACRO-F1 VALUES ARE TEN-SEED MEANS IN PERCENT. THE CONFIRMATORY FAMILY IS RESTRICTED TO A5–A0, A3–A3', AND A5–A6; THEIR REPORTED GAINS USE ONE JOINT FAMILY-WISE NON-STUDENTIZED MAX-ABSOLUTE-CENTERED-DEVIATION HIERARCHICAL PAIRED BOOTSTRAP. OTHER DIRECT CONTRASTS ARE EXPLORATORY AND USE MARGINAL HIERARCHICAL PAIRED-BOOTSTRAP INTERVALS.

ID	Fixed-protocol role or added component	Hard macro-F1	Gain [95% CI]
A0	direct spectral backbone	–	A5–A0: pending formal v2 artifact
A1	single learned mask	–	A5–A1 exploratory: – [–,–]
A2	tri-mask and bypass, CE only	–	–
A3'	A2 + proportional teacher (q_s, q_j, q_u)	–	A3–A3': pending formal v2 artifact
A3	A2 + fixed margin teacher	–	A3–A2 exploratory: – [–,–]
A4	A3 + jammer/quality/orthogonality bundle	–	A4–A3 exploratory: – [–,–]
A5	A4 + exact-source contrast; full VIMD-Net	–	A5–A4 exploratory: – [–,–]
A6	full dual-route composite control	–	A5–A6: – [–,–]
A7	A5 without the applied bypass	–	A5–A7 exploratory: – [–,–]

TABLE V

HARD-INTERFERENCE HEADLINE AMC RESULTS. ACC./F1/WCR ARE PERCENTAGES; LOWER NLL/ECE IS BETTER.

Model	Acc.	F1	WCR	NLL	ECE
Direct backbone	–	–	–	–	–
MCLDNN reimpl.	–	–	–	–	–
IQFormer-inspired	–	–	–	–	–
CSSL-AMC sup. adapt.	–	–	–	–	–
VIMD-Net	–	–	–	–	–

TABLE VI

FACTOR-ISOLATED MACRO-F1 PERCENTAGES AND PAIRED A5-MINUS-REFERENCE GAIN IN PERCENTAGE POINTS. A0 IS THE REFERENCE FOR HARD, OOD, AND RECEIVER ROWS; THE PREDECLARED CSSL ADAPTATION IS THE REFERENCE FOR CLEAN RETENTION. EACH OOD OR RECEIVER AXIS IS INTERPRETED THROUGH ITS PREDECLARED A0-DEGRADATION-CALIBRATED CONFIDENCE-INTERVAL BRANCH; COMBINED OOD IS EXPLORATORY.

Regime	Reference	VIMD-Net	Gain [95% CI]
Hard SIR ≤ 0 dB (A0)	–	–	– [–,–]
Unseen jammer (A0)	–	–	– [–,–]
Unseen speed (A0)	–	–	– [–,–]
TDL-B/E (A0)	–	–	– [–,–]
Combined OOD (A0)	–	–	– [–,–]
Clean A/C/D (CSSL)	–	–	– [–,–]
Clean B/E (CSSL)	–	–	– [–,–]
ADC 10-bit + AGC	–	– pending formal v2 artifact	
ADC 12-bit + AGC	–	– pending formal v2 artifact	
CFO/timing (A0)	–	– pending formal v2 artifact	

VIII. REPRODUCIBILITY AND SCOPE

Each executed run is required to store the complete configuration, environment record, dataset manifest, digest, model seed, checkpoint, source-aligned probabilities, per-regime metrics, and paired statistics. Student initialization occurs after seeding. Validation selection is delayed until the full objective is active. Tables are populated from eligible machine-readable artifacts; the paper has no manually typed performance cells.

a) Scope.: The evidence target is single-antenna AMC under source-disjoint, structured-interference, TR 38.901 TDL-profile simulations. It establishes neither waveform reconstruction nor source separation, open-set or protocol identification, modulation recommendation, complete V2X geometry, SDR/field transfer, or real-time deployment. Unanchored in-taxonomy cochannel and multi-jammer mixtures remain separate tasks under the definitions above.

TABLE VII

MECHANISM AND COMPLEXITY OUTCOMES.

Quantity	VIMD-Net
Teacher–student mask JS $\downarrow$	–
Third-route weighted correlation $\uparrow$	–
Target-energy transfer ratio α	–
Target amplification share $\Pr(\alpha > 1)$ (%)	–
Jammer leakage $\beta \downarrow$	–
Third-route Spearman; permutation p	–; –
Occupancy/gain Spearman; one-sided p	pending v2
Oracle-conditioned spectral ratio gain (dB)	–
Parameters	–
Batch-one P50/P95 on – (ms)	–/–

b) Structural limitations.: First, the teacher requires clean component bookkeeping during simulation training and cannot supervise unreferenced field captures. Second, TDL channels are standards-aligned primitives rather than a complete nonstationary V2X geometry. Third, separately computed component STFT powers omit coherent cross terms and therefore define a simulation-component allocation target, not a conserved decomposition of mixture energy. The masks allocate task representations and do not reconstruct isolated time-domain sources; the additive bypass makes this distinction explicit.

IX. CONCLUSION

This paper formulates interference-robust AMC under vehicular-Doppler TDL-profile simulations as physically supervised representation allocation. VIMD-Net combines a target-power-dominant/jammer-power-dominant/ unexplained-or-power-ambiguous teacher, dual task branches, and exact-source cross-condition learning on a shared complex-STFT lattice. Its simulation claims are deliberately falsifiable through source-disjoint factor holds, single-view inference with source-paired statistics, and oracle-conditioned diagnostic component probes. The internal build establishes the method and locked protocol; quantitative conclusions will be stated only after the predeclared multi-seed evidence gate is satisfied.

INTERNAL SUBMISSION NOTE

Before submission, the human authors must (i) replace authorship placeholders, (ii) verify every scientific claim and

citation against the primary source, (iii) supply eligible locked simulation results without expanding the scope, and (iv) confirm compliance with the live IEEE/TVT policy before setting `\internalreviewfalse`. This internal engineering draft must not be submitted unchanged: an AI tool cannot replace an author in generating article content or assume author responsibility. Human authors must substantively author, independently verify, and take responsibility for the final article. Any permitted AI-assisted modification of text written by the authors must be disclosed in the public manuscript's Acknowledgment. This internal note is an operational gate, not that disclosure.

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